

## **Conclusion /inference**

In this lab, we cleaned the messy air quality dataset. We filled all the missing data using median values and capped the crazy high AQI numbers at 500 because the CPCB scale only goes up to 500. After plotting the distribution, we saw that most cities have moderate pollution, but a few super dirty days pull up the average AQI. So, median AQI is much better for public reporting than the mean.

## Lab 2

### Lab Exercise II: Data exploration and inferences

#### Aim

- To investigate whether worsening air quality across Indian states is linked to declining agricultural output by independently choosing, applying, and justifying appropriate data analysis and visualisation techniques on raw, messy, real-world datasets.

#### Task 6: Is India's air getting better or worse over time?

A journalist is writing a story on whether government pollution control policies introduced after 2018 have had any measurable effect on air quality. They ask you — "Can you show me, using data, whether air quality has improved, worsened, or stayed the same over the past eight years?"

Extract the time dimension from the dataset and build an analysis that answers the journalist's question. Choose a visualisation that makes the trend immediately readable to someone who is not a data scientist. Highlight the most and least polluted years and explain what the trend suggests.

Your plot with the trend clearly visible and key years highlighted

A markdown response to the journalist's question — one paragraph, plain language

*Think: What needs to happen to the Date column before you can group by year? Which plot type communicates a trend over time most clearly? 2 marks*

#### Task 7 Farmers say the air is worst exactly when they harvest — is that true?

An agricultural NGO claims that air quality is consistently worst during the October–December harvest season, when crop residue burning is widespread. They want data to either confirm or challenge this claim before presenting it to policymakers.

Investigate whether AQI follows a seasonal pattern across the year. Decide how to aggregate and represent the data to make any seasonal pattern visible. Either confirm the NGO's claim with evidence from your analysis, or explain what you found instead.

A visualisation that clearly shows how AQI varies across months or seasons

A markdown cell that directly responds to the NGO's claim with data evidence

*Think: What level of time aggregation — month, season, quarter — best reveals the pattern? How do you extract that from a date column? 2 marks*

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**Task 8: Can the two datasets talk to each other?**

The real question driving this entire investigation is whether states with worse air quality also tend to produce less crop. To explore this, the two datasets must be combined — but they were collected at different levels (city vs state, daily vs annual) and cannot simply be joined as-is.

Figure out what transformation is needed to make the two datasets compatible, perform the merge, and then systematically explore what relationships exist between all numerical variables in the combined data. Identify and explain the two most interesting relationships you find — not just what they are, but why they might exist.

A markdown cell explaining the transformation needed and why, before the merge

A visualisation of relationships across all numerical features

Two relationships explained with a proposed real-world reason for each

*Think: If one file has city-day rows and the other has state-year rows, what needs to happen before they can be joined? What does a correlation matrix actually tell you — and what doesn't it tell you? 3 marks*

**Task 9 : The minister needs to act — what do you tell her?**

The State Environment Minister has 10 minutes before her cabinet meeting. She has never opened a Jupyter notebook. Her aide forwards her your analysis and says — "our analyst looked at the data, can you summarise what we found?" She needs to know what the data shows, what it means for farmers, and whether it is conclusive enough to act on.

Write a clear, honest briefing addressed directly to the minister. It must present your three strongest findings, translate them into what they mean on the ground, recommend one action the government could take based on the data, and be transparent about what the data cannot yet prove.

150–200 words in a markdown cell, addressed to the minister. Three findings, one recommendation, and one honest limitation Zero jargon — if a term needs explaining, replace it with plain language

*Think: Which of your findings actually matters to a farmer or a policymaker? What is the difference between correlation and proof? Would the minister act on what you found? 3 marks*

**Optional — advanced task****Build the case — or break it**

Attempt only after completing Tasks 1–9. The central hypothesis of this entire investigation is: states with higher pollution have lower crop yields. Your job here is to either build the strongest possible case for this hypothesis — or find evidence that challenges it.

### **Task A: The two extremes — do they tell the same story?**

If the hypothesis is true, you would expect the most polluted states and the least polluted states to show a clear difference in crop output. But the data might surprise you.

Identify the states that represent the two extremes of pollution. Compare their agricultural output using a visualisation of your choice. Analyse what you see — does the pattern support the hypothesis, contradict it, or is it more complicated than that?

### **Task B: Put a number on the relationship**

A research team wants to know not just whether a relationship exists between AQI and crop production, but how strong it is and in which direction. They will use your output to decide whether to fund a full causal study.

Quantify the relationship between average state AQI and average crop production. Choose a method that both measures the strength and makes the pattern visible in a single output. Interpret your result for the research team — be honest about what it proves and what it does not.

*Think: What does a correlation value of 0.2 vs 0.8 mean in practice? Does a strong correlation mean pollution is causing yield loss?*

### **Task C: One plot to rule them all**

You have produced many visualisations across this lab. A data journalism outlet wants to publish just one chart from your work to accompany a story on India's air-agriculture crisis.

Choose the single visualisation from your entire lab that you believe most powerfully tells the story of this dataset. Write a 3–4 sentence caption for it — not a description of what the chart shows, but an explanation of why it matters and what it reveals that no other chart in your analysis does.

Caption reveals insight, not just description

Across all optional tasks, your writing must acknowledge that correlation is not causation and name at least one factor — rainfall, irrigation, soil type, economic conditions — that this dataset cannot account for but which could explain any relationship observed.

Deliverables

Single Jupyter Notebook (.ipynb) — tasks numbered, code commented, every decision explained in a markdown cell

Minimum 5 visualisations of your own choosing — axes labelled, titles present

Markdown reasoning cells for every task — not optional, part of the marks

## Submission Guidelines

7. Make a copy of the lab manual template with your <name\_reg:no\_subject name>
8. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
9. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
10. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 ...> and push the code to Git.
11. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
12. Upload the PDF to Google Classroom before the deadline.

## Code with Results